

THE INTERNET OF THINGS



News, views and development on the Internet of Things

“Designers need to pay special attention to security-related aspects of any IoT-compatible product”

Kaivan Karimi is the executive director—global strategy and business development for the Microcontroller group at Freescale Semiconductor. In this role, he is responsible for defining and driving the technology, product and business strategies related to the Internet of Things (IoT). Here he speaks to Janani Gopalakrishnan Vikram about protocols, design considerations and tools for the IoT schema

What are the various protocols that are fuelling the growth of the IoT?

There are lots of them, and IoT will be forcing protocol co-existence and harmonisation. A lot of these protocols have to do with the communications technologies that will be used for short- and long-range communications, and the security blocks that need to go along with them. Some of the popular protocols are:

1. Wired communications: IEEE 802.3—Ethernet; CAN; and various power line communication (PLC) standards such as Prime, G3, IEEE 1901 (HomePlug AV) and CENELEC
2. Wireless communications (BAN, PAN, LAN): IEEE 802.15.4 and its associated upper-layer networking protocols such as ZigBee, ISA100.11a, WirelessHART, MiWi and specifically 6LoWPAN, which works with standard Internet protocols; NFC-ISO/IEC 18092; IEEE 802.11 (Wi-Fi); KNX; M-Bus; ANT & ANT+; Bluetooth and BTLE; and EnOcean
3. Wireless communications (WAN): Dash7, Weightless, SIGFOX and IEEE 802.16 (WiMax)
4. Cellular: GPRS/2G/3G/4G

What are the special design considerations that engineers have to keep in mind when developing IoT-compatible products?

A lot of use cases are for battery-operated devices, so low power and long battery life is a key. Choosing the right MCU/MPU, sensors, and communica-

tions technology engine to future-proof the product is important. Finally, security will be extremely important for IoT, and designers need to pay special attention to security-related aspects of any product.

Are there any special components (for communication, security, etc) used in IoT products? What IP cores do you offer for the same?

We are one of the few companies that have product offerings from the edge of the network via sensors (Xtrinsic sensors), MCUs (Kinetis MCUs) and MPUs (i.MX MPUs), all the way through the core of the network via our communications processors (QorIQ). We also offer low-power connectivity solutions for the edge of the network.

We also believe that the role of software is extremely important for IoT-related system development, as the IoT is all about new classes of services that will be offered. In order to spur innovation, partners from the ecosystem need to work together using open platforms, and that is why for our MCUs and apps processors we have standardised on ARM-based cores and products around them.

Could you tell us about Freescale's solutions for the IoT?

Freescale's Tower System is certainly a good way to work with our IoT-related products. It is a modular, scalable and flexible development board for

automotive, consumer, industrial, medical and networking markets. It connects through the serial module that features Ethernet, high-speed USB, RS232/485, CAN and LIN. It spans across 8-, 16- and 32-bit microcontrollers/microprocessors, analogue products, sensors and our wireless connectivity solutions.

We have a lot of products that are suitable for IoT-related applications, and IoT requirements are now part of the broad requirements that each product manager considers. One example is the world's smallest ARM-powered MCU that we announced a few weeks ago. It is amazingly small and low power consuming while retaining rich MCU features.

This product is not only available through our Tower system but also our Freedom Development Platform, which is an ultra-low-cost development platform. Features of our Freedom platform include easy access to MCU input/output (I/O), battery-ready, low-power operation, a standard-based form factor with expansion board options, and a built-in debug interface for Flash programming and run-control.

What toolkits do you suggest for IoT application software development?

Tower and Freedom platforms come with all the software tools and development environment needed, both from Freescale as well as our ecosystem partners.